Fiscal Year:	FY 2009	**Task Last Updated:**	FY 04/01/2009
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Project Title:	Direct Computational Simulations and Experiments for Internal Condensing Flows System-Instabilities/Dynamics in Microgravity and Terrestrial Environments		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	10/01/2004	**End Date:**	09/30/2010
No. of Post Docs:		**No. of PhD Degrees:**	4
No. of PhD Candidates:	4	**No. of Master' Degrees:**	3
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2010 per PI (2/2010) NOTE: End date changed to 01/01/2009 per PI and T. Rusnak/GRC (10/08) NOTE: 8/06--funding reinstated until 9/30/2007, per J. Cohen (HQ)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GB52G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Understanding of qualitative phenomena based on experiments supported by accurate computational simulations (steady and unsteady) for quasi-steady annular/stratified internal condensing flows in micro-gravity and terrestrial environments is essential for good design and performance of condensers in applications such as Rankine-cycle based power (nuclear) plants, Thermal Management systems, Looped Heat Pipes, Capillary Pumped Loops, etc.
Condenser-sections in these devices often achieve complete or incomplete condensation by the exit under unspecified or specified (as being proposed) exit conditions (which are exit vapor quality or liquid fraction or pressure) while the inlet (pressure and flow rate) and wall conditions are typically always known or specified. These condensing flows' attainability/stability of steady flows, intrinsic waviness, noise-sensitivity (primarily due to minuscule transverse vibrations of the condensing surface), and exit-condition sensitivity critically depend on the presence or absence of gravity in driving the condensate.

Zero-gravity simulations establish that the parameter zone within which steady flows exist for any space-based condenser, experiencing full or partial condensation with unspecified (free) exit condition (as this is a typical exit condition for condensers in many thermal systems), is very small to non-existent as the flows are unstable and will peel off from the surface. Therefore quasi-steady flows in zero- or micro-gravity under unspecified exit pressure conditions (for situations where only the heat-transfer surface temperature is known) is actually impossible. This fact is not known as operation of condensers (or boilers) in this mode is often proposed for many space-based thermal system designs. Thus achieving quasi-steady flows in such space-based condenser employments requires flow control strategies suggested by this research. This strategy stabilizes the flow by fixing the mechanical/dissipative energy in the condenser - just as an upside down pendulum is stabilized by controlling the base motion to fix its energy levels.

The ground based experiments and supporting simulations (both for terrestrial and micro-gravity environments) are to explore how exit-condition specifications based flow control strategies can used to attain different quasi-steady flows and avoid unwanted unsteady/oscillatory flows. Without the recommended control, the unsteady/oscillatory flows may arise from changes in flow regime within the condenser or from system-instabilities (due to interaction of condenser with other upstream or downstream components of the system).

Although the proposed research involves fundamental investigations of flows inside relatively simpler tube and channel geometries, the qualitative understandings/results and tools that will be obtained are going to be sufficiently fundamental and general in nature that they will provide an important framework for facilitating the design, incorporation, and operation of new realistic space-based or ground-based condensers (that is condensers that employ more complex but realistic geometries such as SWEP/Tranter plate heat-exchangers, shell-and-tube type, etc.) in planned systems. In fact the basic result regarding a need for independent pressure-difference imposition (or exit pressure specification if inlet pressure is known) is general enough that it applies to flow-boilers as well.

Task Objectives

Experimental

• Complete experimental investigation of attainability/stability and exit-condition issues for downward and steady internal condensing flows inside a vertical tube. Following this gravity driven condensate flow investigations, initiate similar experimental investigations of gravity-to-shear driven internal condensing flows in inclined-to-horizontal channel (i.e. rectangular cross-section of suitable aspect-ratio) geometries which experience condensation on its bottom wall.

• Ensure that the ground based experimental facility is well instrumented and progressively capable of providing more and more detailed information that can be synthesized with results obtained from the computational capability being developed for these flows.

• Complete ongoing invention efforts and develop a non-intrusive film thickness sensor (based on principles of fluorescence and fiber-optics) that is capable of measuring time-varying thickness of the dynamic condensate. Verify such an invention by independent thickness measurements. Make progress on employing such sensors for the gravity driven annular flow cases in a vertical tube geometry and/or inclined-to-horizontal channel (i.e. rectangular cross-section of suitable aspect-ratio) geometry that experiences condensation on its bottom wall.

Computational

• Complete development of accurate simulation tool for laminar vapor/laminar condensate internal condensing flows for channel or vertical tube geometries.

• Ensure that the simulation tool accurately locates the interface (under smooth or wavy conditions) for terrestrial and zero-gravity environments and meets various convergence and grid independence criteria for both specified and unspecified exit condition cases.

• Ensure that the simulation tool is capable of reliably reproducing known results for classical benchmark problems (such as Nusselt problem of film condensation on a vertical plate, forced vapor flow experiencing film condensation on a horizontal plate, etc.) and, also, capable of providing new non-linear stability results consistent with known experimental results for these classical problems.

• The verifications of the two dimensional code on classical problems as well as through successful comparisons of the simulation results for the ongoing ground based experiments are necessary for further enhancements (compressibility effects under specified exit conditions, three dimensional simulations, etc.) of the code. The verifications will also make simulation tool based zero gravity predictions more trustworthy.

• Eventually the simulation tool should be reliable enough to help plan proper space based experiments and, also, to help reduce the number of design iterations needed to have a condenser (or boiler) operate successfully in a zero gravity environment.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:	Issue-1: Can one achieve, in zero or micro-gravity, an internal condensing flow in which the condensate adheres to the condensing surface? Can one do this even if the natural tendency of the condensate is to peel off from the condensing surface? Can an approach be developed to make the condensate adhere to the condensing surface? This is desirable for ensuring efficiency and functionality of space-based condensers. Research Result: Yes, such flows can be achieved, but only with the help of the recommended flow control. The simulation results show that annular/stratified condensing flows in zero-gravity are unstable under ordinary arrangements for the operation of the condenser. These results indicate that, for zero gravity, without the recommended flow control, the natural tendency of the condensate is to fly off the condensing surface. Both the natural instability and the stabilizing control strategy results are described in this final report. This research provides reliable quantitative predictions and control strategies for imposing/fixing the exit value of the condensate flow rate which leads to stabilization of an annular/stratified flow in zero-gravity conditions. We recommend that a condenser in space be made of two coupled condensers – one involving a controlled partially condensing flow (say up to 80% condensation) and the other involving uncontrolled condensation of the remaining vapor. Thus the recommended approach enables achievement of robust annular/stratified flows that are necessary to maintain high thermal efficiency of space-based condensers. Issue-2: Can ground-based experiments and relevant simulations/theory for gravity driven as well as shear driven cases be synthesized together? Can one use such experimentally validated simulations and theory to demonstrate the feasibility of attainment and control of robust annular/stratified flows in space-based condensers? Research Result: Yes, results obtained from the simulation tools show agreement with the key hypotheses, relevant carefully conducted ground-based experiments, and other well known facts/solutions for condensing flows. Since the reported steady and unsteady simulation results for partially and fully condensing flows in zero-gravity are obtained from two independent tools (a full two-dimensional CFD code and a quasi one-dimensional code), the recommended flow control strategy is expected to lead to flow stabilization. Thus the reported zero-gravity simulation capability and results make both qualitative and quantitative predictions that are trustworthy aids in the design of space-based and 0g experiments. As mentioned above, the research done so far addresses the primary concerns of NASA. However, additional experimental/computational research is required to make the existing 0g/shear driven simulation results usable for design of zero-gravity experiments. This report also describes how the novel principles and experimental approaches reported here can be strengthened by such additional experiments and simulations. This additional research will mark a path that will lead to successful development/facilitation of various technologies such as space based Rankine cycle [1] or related thermal management systems [2] that employ condensers. Software: The ongoing/proposed simulation tool component will have a far reaching impact as it will shed light on the physics of an array of related problems (viz. boiling, condensing, and evaporating phase-change flows of pure fluids) – both in simple geometries of fundamental interest and complex geometries of engineering interest. Furthermore, eventual inclusion of diffusion equations (available through COMSOL's CFD tools) in the simulation tool can allow one to model diffusion processes in vapor and/or liquid phases. Examples of innovative research in the area of phase-change flows involving either of the two types of diffusion are: condensation in the presence of non-condensable gas and evaporation of saline liquid films. Both these simulation abilities will assist in improving technologies related to cleaning/recycling of air and water for long term human presence in reduced and micro-gravity environments. Earth Benefits The exit condition based flow categorizations and experimental results are also going to be useful for many new terrestrial systems (such as two-phase electronic or chip cooling) that need to employ a condenser with a steady and stable flow for the thermal system. While it is true that many terrestrial systems can be made to function by sufficient number of trial and error experiments or by an on-off strategy that does not require attainment of steady flows, the ground based results from this research will help significantly in innovation and improvement of superior new technologies for such systems. For example, a specified exit condition mode of operation (based on electronic flow control) is expected to allow steady and stable operation of many terrestrial condensers in novel systems (electronic cooling, etc.) with the ability to handle a larger range of heat-load variations while cutting down on the system's start up times. Also, the development (nearly complete) and deployment of film thickness sensor will be valuable in the measurement processes that are important to the development of many terrestrial applications that involve film cooling, film evaporation, film flows in chemical process industries, etc.
Task Progress:	Hypothesis-1/Result-1: Quasi-steady internal condensing flow behavior is expected to be "elliptic" which means that different quasi-steady vapor-liquid configurations – particularly those associated with annular/stratified flows are possible for the same values of inlet pressure, inlet mass flow rate, and wall temperature conditions. These different quasi-steady flows for which the liquid adheres to the condensing surface - can be achieved by controlling and steadying the mechanical power input to the condenser. This is possible through imposition of appropriate quasi-steady exit conditions involving fixing of the liquid fraction (or vapor quality) at the exit of a partially condensing flow, and, possibly with less efficiency, pressure difference across fully condensing flows. Hypothesis-2/Result-2: Despite Hypothesis-1, the strictly steady equations that govern annular/stratified internal condensing flows are shown to be "quasi-parabolic" because a unique exit condition (exit liquid fraction or exit vapor quality or exit pressure) may be expected as a part of a unique steady solution under self-selected exit conditions. These strictly steady equations are termed "quasi-parabolic" because, unlike "strictly parabolic" incompressible single-phase or air-water flows, they do not "strongly" self-select their exit condition under all conditions (1g and 0g) and also they may not reject alteration in the interior of the flow because of changes in exit condition. As a result of this special flow physics feature, depending on flexibility in the flow-loop arrangements, these flows may achieve/attain multiple nearby quasi-steady flows for multiple exit conditions. Hypothesis-3/Result-3: For internal condensing flows, the unique steady solution associated with the strictly steady equations can be stable or unstable in nature. This stability is reflected by the will by which any solution, in an unsteady mode, gets attracted to the associated unique steady solution when no exit conditions are prescribed and the flow is allowed to seek its unique exit pressure. This attractor is very strong in case of gravity driven flows and is very weak in case of 0g flows.

	Hypothesis-4/Result-4: Regardless of whether the underlying annular/stratified steady solution is unstable or stable, one is able to stabilize (or make it more robust) an annular/stratified quasi-steady condensing flow by fixing the mechanical power input to the condenser in a flow-loop arrangement. For partial condensation, this can be done by controlled fixing of the exit vapor quality or liquid fraction. As a result of this type of fixing of mechanical or dissipative power within the condenser, an unstable zero-gravity annular/stratified condensing flow can be stabilized and realized. This is equivalent to stabilizing an upside down pendulum through well known control strategies that fix the pendulum's energy level.
Bibliography Type:	Description: (Last Updated: 06/28/2013)
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